PediVAS/Centrimag VAD troubleshooting reference guide

Medical team to resuscitate patient while ECMO specialist troubleshoots VAD – KIV increase vent, BVM, Adrenaline, CPR

Issue	Management steps	Key learning points/differences from ECMO
Low flow/ line chattering	- Evaluate PATIENT and treat accordingly (treat patient, not VAD) - check BP, HR, CVP, NIRS, UOP, lactate - check for kinking in cannulas - replace volume and monitor response (*avoid excessive unnecessary fluid boluses – RV may not tolerate. Consider other causes if chattering ++ without obvious source of fluid/blood loss) - consider decreasing RPM (*avoid rapid flow changes) <u>Potential causes:</u> - hypovolemia (low CVP) - **RV failure** (high CVP - KIV iNO, RV support) - tamponade (high CVP) - tension pneumothorax - cannula kinking/obstruction (eg massive thrombus) - cannula positioning/suctioning of cannula on ventricle wall - high afterload (KIV increase sedation, afterload reduction)	VADs are preload dependent and afterload sensitive. Flow target range should be discussed daily with VAD team and documented. ECMO specialists and primary team can change RPMs as part of our routine troubleshooting if necessary. (RPM should be > 1000). Should still target for flow within target range. Unlike VA ECMO, change in flow on LVAD have implications on LV filling, septal wall positioning, and ability of RV to cope with changes. Closely monitor CVP and markers of perfusion. VAD team should be alerted if change in **<u>flow</u>** outside target range. Echo may be needed to evaluate RV function, LV filling, septal wall positioning, volume load on heart/ventricular size, pericardial effusion, clot, cannula positioning.
Console or motor failure	1) Medical team to bedside to support/resuscitate patient 2) Turn on back up console and make sure new motor head is close enough to patient for exchange 3) Turn RPM down to 1000 (if motor has completely stopped working, skip to step 4) 4) Clamp **outflow cannula** (VAD TO patient) *refer to Fig 1 and 2 5) Turn RPM to ZERO 6) Unscrew pump lock and remove pump 7) Insert pump into new spare motor and lock into place 8) Secure flow sensor for back up motor onto **outflow cannula** 8) Increase RPM to 1000 9) Unclamp cannula 9) Increase RPM to baseline	Pump change should be initiated by bedside team with medical team at bedside to support patient, no need to wait for perfusionists.

Issue	Management steps	Key learning points/differences from ECMO
Noise from pump	Same steps as above for " (1) Console or motor failure" but to reposition and insert pump back onto same motor. (no need for back up motor)	Pump repositioning should be initiated by bedside team with medical team at bedside to support patient, no need to wait for perfusionists. Noise from pump is most commonly due to malposition of the pump head on the motor leading to rubbing on one portion of the motor. May also alarm “Pump Not Inserted”.
Air bubble in circuit (in either cannula or in pump) **Massive/mobile clot in circuit (in either cannula or in pump)** **Tubing disconnection, rupture or cannula dislodgement**	1) Medical team to bedside to support/resuscitate patient 2) Immediately clamp **outflow cannula** (VAD TO patient) **URGENTLY** 3) Clamp **inflow cannula** (FROM patient to VAD) *Refer to fig 1 and 2 4) Prepare back up **pre-primed ECMO circuit** for circuit change	Unlike ECMO, no oxygenator to catch air bubble/clot. Therefore air or mobile clots anywhere in circuit, including **inflow cannula** (FROM patient to VAD) should also be addressed urgently. In an emergency/eCPR, initiate circuit on clear prime and transfuse PCT to patient immediately after.
CPR	- Intubate patient - If this is a circuit-related emergency -clamp cannulas. - If no issues with circuit (eg collapse due to RV failure), no need to clamp cannulas - Conventional CPR for closed chest. Open chest CPR as per our institutional policy - KIV Conversion to VA ECMO	Risks of CPR are cannula dislodgment, rupture of fragile tissue, hemopericardium. However, if indicated team should proceed with CPR and resuscitation. If CPR is needed and no issues with circuit/VAD, do not clamp circuit and resuscitate through VAD. As discussed with CTS, if indicated to proceed with CPR with conventional positioning of hands on chest. Echo and CXR following ROSC given potential complications of CPR with cannulas in situ.
Synchronized Cardioversion/ Defibrillation	- No changes in practice for defibrillation. Pad placement as per normal protocol, may be limited by dressing/open chest - No need to remove pump from motor, continue to let VAD run - Ensure that machine and pump/motor are away from patient bed. Avoid pads on cannulas - Confirm pump flows, rpm and alarms immediately after the shock.	

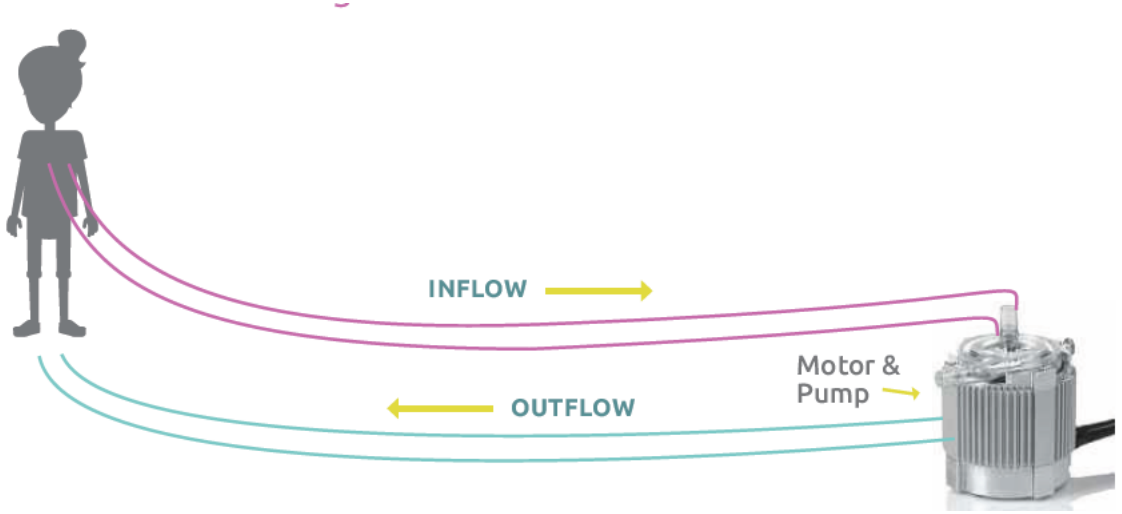


Fig 1: Naming of cannulas are relative to VAD. Inflow = patient to VAD, outflow = VAD to patient



Fig 2: Clamping on Berlin cannulas –clamps perpendicular to cannulas, close to patient